Zhi Wang

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Education

University of California, San Diego — San Diego, CA, USA
Ph.D. in Data Science — *Sept. 2025 – Ongoing*

University of Science and Technology of China — Hefei, China
B.S. in Mathematics, School of Gifted Young — *Sept. 2021 – Jun. 2025*

University of California, Berkeley — Berkeley, CA, USA
Visiting Undergraduate Student — *Jan. 2023 – Dec. 2023*

Research Interests

- **Reliable agentic systems:** monitoring, steering, and credit assignment for agents, and the infrastructure for scientists to collaborate with them.
- **Predictive science of optimization:** theory and forecasting of training dynamics, e.g. adaptive optimizers and predicting or monitoring learning metrics.
- **Embodied agents:** orchestrating reasoning and low-latency control for robotic tasks across different horizons.

Publications

- Ke Zhang, Patricio Gallardo, Sudhir Murthy, Yi Xie, **Zhi Wang**, and Maziar Raissi, *Beyond Compilation: Human-Calibrated Evaluation of Semantic Faithfulness in Lean Autoformalization*, NeurIPS 2026 MATH-AI Workshop. [Paper]
- Eon Lee, Andrés R. Vindas-Meléndez, and **Zhi Wang**[1], *Generalized Snake Posets, Order Polytopes, and Lattice-Point Enumeration*, Discrete Mathematics (2026): 115072.

Open Source Contributions

RCP: Research Control Platform

- Harness and control plane over coding agents: one platform for research artifacts and asynchronous compute jobs, mediating how agents interact with humans and with compute.
- Conversational research assistant as the unified entry point: requests like "start the experiment for me" become managed asynchronous jobs, with human approval gating protected actions.
- Git-like research graph of questions, hypotheses, experiments, and evidence, with branching and merging.
- Runs as a macOS app or a self-hosted Linux team server with multi-member collaboration and attribution; agent lifecycles are verified by a TLA+-style state-space explorer.

[1] In mathematics, authors are conventionally listed alphabetically, signifying equal contribution.

Terminal-Bench

- Contributed two benchmark tasks: merge-diff-arc-agi, included in Terminal-Bench 2.0, and solve-sudoku.

Current Research

- Comparing two ways to train language-model theorem provers in Lean: writing the whole proof and fixing it from verifier feedback, or proving step by step with tree search and expert iteration, starting from the same checkpoint with compute held equal.
- Studying how sequential, mixed-reward, consolidation-based and stabilized reinforcement learning trade off keeping skills in trained domains against staying able to learn new ones.
- Predicting and steering away from chain-of-thought loops in reasoning language models using their internal activations, and studying why lightweight activation edits make models like Olmo-3 reason better on math.
- Making vision-language model agents faster and cheaper at robot manipulation, both by controlling the robot directly and by instructing a vision-language-action policy.
- Extending self on-policy distillation to multi-turn agentic settings, testing whether its denser training signal yields better small SWE agents.

Research Experience (Undergraduate)

Ehrhart Theory of Order Polytopes *Jun. 2023 – Sept. 2024*
Mentor: Andrés R. Vindas Meléndez, UC Berkeley

- Proved closed-form and combinatorial formulas for a family of order polytopes, establishing key base cases and monotonicity results by introducing a novel lattice path coloring method to connect geometric objects with combinatorial enumerations.

Accelerating Grokking via Low-Rank Structures *May. 2024 – Oct. 2024*
Mentor: Difan Zou, University of Hong Kong

- Investigated the NTK-to-rich regime transition, leveraging low-rank initialization in MLPs and transformers to accelerate grokking for various tasks.

Simulate Worm Neural activity *Sept. 2024 – Jun. 2025*
Mentor: Quan Wen, University of Science and Technology of China

- Trained RNN simulators based on C. elegans connectome to reproduce neural activities, carried out ablation study and visualization to guide training principles for neural simulators.

Teaching

- Teaching Assistant, DSC 242: High-Dimensional Probability and Statistics, UC San Diego, Fall 2026.
- Teaching Assistant, DSC 140B: Representation Learning, UC San Diego, Winter 2026.